

Andreas Tzitzikas

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Education

University of Maryland, College Park, MD Expected May 2027
Master of Science in Computer Engineering **GPA: 3.9**
Bachelor of Science in Computer Engineering (GPA: 3.89) Expected May 2026
Relevant Coursework: Elements of Discrete Signal Analysis (ENEE222), Introduction to Micro/Nanoelectronics (ENEE304), Microprocessors (ENEE440), Operating Systems (ENEE447), Reverse Eng. & HW Security (ENEE459B)

Experience

Embedded Systems Intern, Alchemity – College Park, MD Jun 2025 – Aug 2025

- Designed mixed-signal control loops for fuel cell reactors, integrating high-precision **ADC** sensors and **PWM** motor drivers with noise-filtering firmware logic.
- Developed embedded firmware with interrupt-driven architectures, real-time scheduling, and robust error handling for industrial control applications.
- Integrated analog sensors and actuators with digital control systems, ensuring signal integrity, noise immunity, and reliable operation in harsh environments.

Undergraduate Teaching Assistant, ENEE205 Electric Circuits – College Park, MD Sep 2025 – Present

- Lead laboratory and discussion sections covering circuit theory fundamentals, Kirchhoff's laws, Thevenin/Norton equivalents, and AC/DC circuit analysis.
- Provide hands-on guidance with oscilloscopes, signal generators, and circuit simulation tools for practical circuit design and measurement techniques.

Technical Coordinator, Electronics Prototyping Lab, Terrapin Works – College Park, MD Jan 2024 – Present

- Maintain and operate **LPKF** PCB prototyping equipment ensuring reliable fabrication capabilities with **90% uptime** for student and faculty projects.
- Provide technical support for PCB design, fabrication, and assembly processes including schematic capture, layout, and component soldering techniques.

Projects

STM32G4 Bare-Metal Peripheral Integration | ARM Assembly, Analog/Digital Interfaces Jan 2025 – May 2025

- Implemented direct register-level programming for **ADC**, **DAC**, **PWM/PFM** generation using **TIM2**, and **SPI** flash memory interfaces on **STM32G4** microcontroller platform.
- Engineered multiplexed interrupt system routing single hardware interrupt to **4 distinct peripheral drivers** with state-based arbitration for efficient resource utilization.
- Developed comprehensive analog signal processing including **ADC** sampling, **DAC** output generation, and **PWM** motor control with precise timing and synchronization requirements.

Dual-Host S2-LP Radio Shield | KiCad, RF Design, Multi-Platform Firmware Jan 2024 – Present

- Designed dual-host development board for **S2-LP** sub-1 GHz transceiver with **STM32** and **RP2350** compatibility, implementing complete **RF** circuit with matching network and crystal oscillator for optimal radio performance.
- Developed shadow mode debugging architecture where **STM32** passively monitors **SPI** traffic between **RP2350** and radio, enabling cross-platform firmware migration and protocol analysis for wireless development.
- Created comprehensive firmware suite for both platforms including **SPI** drivers, radio configuration, packet transmission, and bus monitoring capabilities with detailed technical documentation.

Tomasulo Out-of-Order RISC-V CPU | Digital Design, Timing Analysis, Verification Nov 2025 - Dec 2025

- Architected out-of-order execution system using **Tomasulo's Algorithm** with register renaming, reservation stations, and reorder buffer for advanced digital system design.
- Achieved **36% higher operating frequency (69.7MHz vs 51.2MHz)** and **40-150% IPC improvement** compared to in-order implementation through timing optimization and performance analysis.
- Synthesized to fabrication-ready GDSII using **Sky130 130nm PDK**, achieving **zero DRC/LVS errors** and comprehensive static timing analysis for reliable digital circuit operation.

Skills

Programming Languages: SystemVerilog, Verilog, C/C++, ARM Assembly, Python, Tcl, Bash, Rust, Java, OCaml, CUDA
Circuit Design: Analog/Digital Circuits, Signal Processing, Power Systems, Filter Design
Electronics: Microcontrollers, ADC/DAC, PWM, SPI/I2C/UART, GPIO, Embedded Programming
Laboratory Tools: Oscilloscopes, Signal Generators, Multimeters, PCB Fabrication
Software Tools: EDA Tools (KiCad), Cadence Spectre, MATLAB, LTspice, Xilinx Vivado